

Construction and Validation of a Profile HMM for the Identification of the Kunitz/BPTI Domain



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Abstract

The Kunitz/BPTI domains are small cysteine-rich protein domains involved in serine protease inhibition, venom toxicity, and other biological functions. In this project, a Hidden Markov Model profile was constructed to identify Kunitz-domain proteins using a structure-based alignment of representative Kunitz sequences. Candidate structures were retrieved from the Protein Data Bank. Redundancy was reduced with MMseqs2, and representative structures were aligned with PDBeFold. The HMM construction was performed with HMMER. The resulting profile was searched against positive and negative UniProt/Swiss-Prot datasets and evaluated across E-value thresholds from $1e-1$ to $1e-15$. The final validation used two subsets. An E-value threshold of $1e-5$ was selected as the best operational compromise across both validation subsets.

At this threshold, the model achieved MCC values of 0.9950 and 0.9849 for validation sets 1 and 2, respectively. Semi-log ROC and FP/FN trade-off plots confirmed that the model maintained high sensitivity while keeping the false positive rate very low. Overall, the constructed HMM showed strong performance for Kunitz/BPTI-domain detection in a highly imbalanced validation setting.

1. Introduction

The Kunitz domains are small cysteine-rich protein domains that are widely distributed across natural systems, from microbes to mammals. They are commonly associated with serine protease inhibition, but they can also function as toxins in venomous animals and may participate in other biological activities, including ion-channel modulation. The typical Kunitz motif is approximately 50–70 amino acids long and adopts a conserved fold composed of β -structural elements and short helical regions, stabilized by three conserved disulfide bridges. Although this structural framework is highly constrained, Kunitz-domain proteins show substantial sequence and functional diversity.

Because of this variability their identification is not always straightforward using simple pairwise similarity approaches. For this reason Hidden Markov Models Profiling pipelines are well suited for this type of domain-detection problem because

they capture position-specific conservation patterns, insertions, and deletions from a multiple sequence alignment. In this project, a profile HMM was constructed for the Kunitz/BPTI domain using representative structurally characterized sequences. The model was then searched against positive and negative UniProt/Swiss-Prot datasets and evaluated across different E-value thresholds to determine its ability to distinguish Kunitz-containing proteins from non-Kunitz proteins.

2. Materials and methods

2.1 Structural data acquisition and filtering

A custom report was retrieved from the Protein Data Bank to collect experimentally resolved structures associated with the Kunitz/BPTI domain. The PDB advanced search was performed by using the Pfam protein family identifier PF00014 that corresponds to the Kunitz/BPTI domain. The search was further restricted to structures with a data collection resolution of ≤ 3.5

Å to retain high-quality and domain-sized structural entries. Polymer entity sequence length between 40 and 80 amino acids to match the expected size of an individual Kunitz domain, rather than full-length multidomain proteins.

The resulting custom CSV report included structural identifiers, chain/auth asym identifiers, and the sequence. The CSV file was processed in Python to remove incomplete entries, duplicated sequences, entries without usable sequence information, and entries outside the selected length range.

2.2 Clustering and structural alignment

To reduce redundancy among similar Kunitz-domain sequences, the filtered sequences were clustered using MMseqs2. Both parameters, minimum sequence identity and minimum alignment coverage were set on 0.95. This produced 20 representative sequence clusters, which were used to define a non-redundant set for downstream structural comparison. The representative PDB-chain identifiers were then formatted as list of PDB codes compatible to PDBeFold input requirements and was submitted for multiple structural alignment. The resulting FASTA alignment was used as the input alignment for HMM construction. In addition, the PDBeFold superposition file was visualized in UCSF Chimera to inspect the structural overlap among the selected Kunitz-domain representatives.

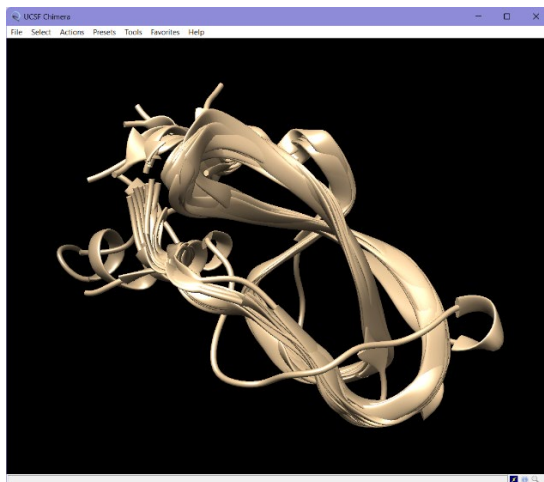


Figure 1. Visualization of superimposition on Chimera

2.3 HMM construction and database search

From the constructed of the structure based alignment we constructed a Hidden Markov Model profile using the HMMER software package locally. The resulting profile was then used to search two separate FASTA datasets downloaded from UniProt/Swiss-Prot. One positive dataset that contained proteins annotated as Kunitz-domain proteins and one negative dataset without Kunitz-domain annotation.

The HMMER searches were performed in sensitive mode, and tabular outputs were generated for downstream processing. The effective database size was fixed during the searches to allow E-value comparability between the positive and negative datasets. From the HMMER output, protein identifiers and domain E-values were extracted and used for subsequent validation.

2.4 Validation dataset preparation

The HMMER table outputs were converted into labelled evaluation files containing three columns: protein identifier, domain E-value, and true class label. Positive proteins were assigned label 1 and negatives 0. The full negative evaluation file was then randomized and divided between two validation subsets. Each validation subset contained 199 positive proteins and 287,115 negative proteins which is half of each.

2.5 Performance evaluation and visualisation

Model performance was evaluated across E-value thresholds from $1e-1$ to $1e-15$. Proteins with domain E-value less than or equal to the threshold were classified as Kunitz-positive at each threshold. While proteins above the threshold were classified as negative. The confusion matrix and derived metrics, including accuracy, Matthews correlation coefficient, sensitivity, specificity, false positive rate, precision, and F1 score were calculated using a Python script.

Because of the highly imbalance between the two validation datasets, the Matthews correlation coefficient was used as the main metric for threshold selection. ROC visualization was also performed in Python using the false

positive rate and true positive rate calculated across multiple thresholds. Since the false positive rate was extremely low, a semi-logarithmic x-axis was used for clearer visualization with Matplotlib. In addition to visualize how the two error types changed across decreasing E-value thresholds and to support the final threshold selection to a false positive/false negative trade-off plot was generated. An HMM sequence logo was generated using Skylign to inspect conserved positions in the final Kunitz profile.

3. Results

3.1 Dataset composition

After filtering the PDB custom report and reducing sequence redundancy with MMseqs2, 20 representative Kunitz-domain clusters were selected for structural alignment. Then the PDBFold alignment of the clusters was used to generate the final alignment input for the profile HMM. Finally the resulting HMM was evaluated against positive and negative UniProt/Swiss-Prot datasets.

1AAL_B 1AAP_A 1D0D_A 1F5R_I 1FAK_I 1G6X_A 1KNT_A 1TFX_D 1YCO_I
1YLC_B 1ZRO_D 2F13_I 2ZJX_B 3FP7_J 3WNY_I 4DTG_K 4ISO_B 4TPI_I
5JB6_C 5YWI_I

Figure 2. The clusters acquired from MMseq2

The positive dataset contained 398 Kunitz-annotated proteins, all of which produced a reported HMMER hit. The negative dataset contained 574,229 proteins without Kunitz annotation. Among these, 6,525 produced a reported HMMER hit, while 567,704 produced no hit and were reinserted into the evaluation file with an artificial E-value of 100. The final negative evaluation dataset therefore preserved all 574,229 negative proteins. Two validation subsets were then constructed, each containing half of the original pools randomized into 199 positive proteins and 287,115 negative proteins for each set.

Table 1. Dataset Components Count

Dataset component	Count
Positive UniProt proteins	398

Negative UniProt proteins	574,229
Positive HMMER hits	398
Negative HMMER hits	6,525
Negative nonmatches	567,704
Total negative evaluation entries	574,229
Validation set 1	287,314
Validation set 2	287,314

3.2 Classification performance at the selected threshold

To identify a threshold that maintained high detection of true Kunitz-domain proteins while limiting false positive predictions among the much larger negative dataset, the model was evaluated on the two validation subsets. The validation used a spectrum across E-value thresholds from 1e-1 to 1e-15.

In the first validation subset the thresholds between 1e-7 and 1e-11 produced perfect classification. However, this behavior was not reproduced in the second validation subset. In the second subset there were stricter thresholds that increased the number of false negatives. Therefore, the threshold was not selected based only on the best result in one validation set. Instead, the performance of both subsets was considered together.

Table 2. Different options for chosen threshold

Thres hold	Set 1 MCC	Set 2 MCC	Mean MCC	Set 1 FP/F N	Set 2 FP/F N
1e-5	0.9950	0.9849	0.9899	2 / 0	2 / 4
1e-6	0.9975	0.9797	0.9886	1 / 0	1 / 7
1e-7	1.0000	0.9797	0.9898	0 / 0	1 / 7

Finally a E-value threshold of 1e-5 was selected as optimal. At this threshold, validation

set 1 produced 199 true positives, 287,113 true negatives, 2 false positives, and 0 false negatives, with an accuracy of 0.999993 and an MCC of 0.995009. Validation set 2 produced 195 true positives, 287,113 true negatives, 2 false positives, and 4 false negatives, with an accuracy of 0.999979 and an MCC of 0.984851. Therefore, the selected threshold provided a strong compromise between sensitivity and specificity. It kept the number of false positives very low in both validation subsets, while avoiding the larger increase in false negatives observed at more stringent thresholds.

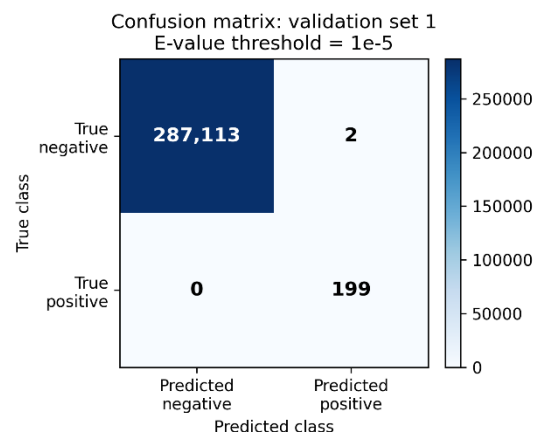
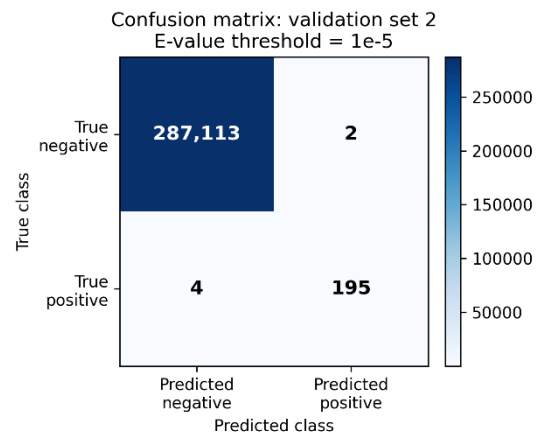
Table 3. Validation set 1 performance at E-value threshold 1e-5.

Metric	Value
Threshold	1e-5
Accuracy	0.9999930389747802
MCC	0.9950089721735175
Sensitivity	1.0000000000000000
Specificity	0.9999930341500792
FPR	6.965849920763457e-06
Precision	0.9900497512437811
F1 score	0.9950000000000000
True negatives	287,113
False positives	2
False negatives	0
True positives	199

Table 4. Validation set 2 performance at E-value threshold 1e-5

Metric	Value
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Threshold	1e-5
Accuracy	0.9999791169243406
MCC	0.9848506143749253
Sensitivity	0.9798994974874372
Specificity	0.9999930341500792
FPR	6.965849920763457e-06
Precision	0.9898477157360406
F1 score	0.9848484848484848
True negatives	287,113
False positives	2
False negatives	4
True positives	195



3.3 ROC curve

The semi-logarithmic ROC curve was generated from the combined validation data to visualize the threshold-dependent behavior of the Kunitz HMM. The curve shows that the model maintained a high true positive rate across the tested thresholds. At the same time the false positive rate remained extremely low. As observed most of the points are concentrated close to the upper-left region of the ROC space. Indicating the strong separation between Kunitz-domain proteins and the negative Swiss-Prot background.

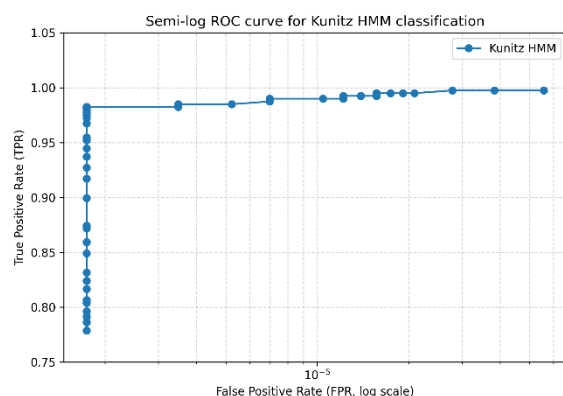


Figure 3. Semi-logarithmic ROC curve for the combined validation dataset.

To visualize how the two error types changed, the false positive/false negative trade-off plot was generated. At more permissive thresholds, the model accepted weaker HMMER hits as positive predictions, increasing the number of false positives. As the threshold became stricter, false positives decreased, but the model began to miss more true Kunitz-domain proteins, increasing the number of false negatives. This inverse relationship illustrates the practical threshold-selection problem, a permissive threshold risks including non-Kunitz proteins, while on the other end overly strict threshold risks excluding divergent or weaker Kunitz-domain hits. The selected threshold is a balance operational cutoff since it was positioned in the transition region where both false positives and false negatives remained low.

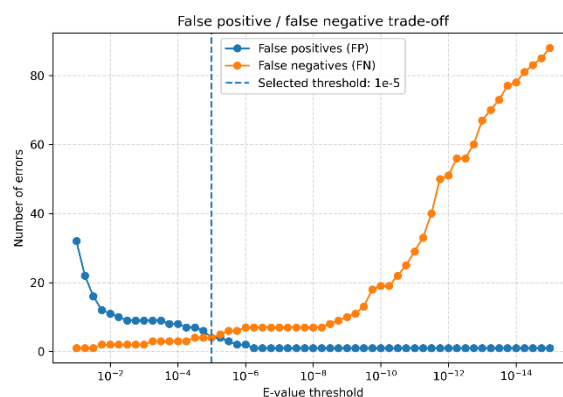


Figure 4.. False positive and false negative trade-off across E-value thresholds. Skylign sequence logo of the final Kunitz profile HMM. selected threshold of 1e-5.

3.4 HMM logo analysis

The Skylign logo was generated to visualize residue conservation in the final Kunitz profile HMM. The logo showed a combination of highly conserved positions and more variable regions across the alignment. In this profile we can see the characteristic conserved cysteine positions that supports the presence of the disulfide-stabilized Kunitz scaffold. Other conserved residues include aromatic and hydrophobic positions that could contribute to the structural stability of the domain. This pattern is consistent with the Kunitz/BPTI domain, the logo further supports that the alignment used for the HMM construction retained the domain sequence signals while at the same time representing the diversity of the selected sequences. In conclusion the overall fold is conserved but sequence variation is tolerated at many positions.

4. Discussion and Conclusion

The HMM profile constructed in this project showed strong ability to identify the Kunitz domain in a highly imbalanced validation set. The model maintained high sensitivity and produced very few false positives when tested against a more than half a million proteins in the Swiss-Prot database. This is an indication that the structural alignment used for the HMM construction successfully captured the relevant conserved features of the domain.

The selected E-value threshold of 1e-5 represented a practical balance between sensitivity and specificity. More permissive thresholds increased the number of false

positives, while stricter thresholds reduced false positives but caused an increase in false negatives, especially in the second validation subset. These false negatives may correspond to more divergent Kunitz-domain proteins, fragments, or proteins whose sequences differ from the conservation pattern encoded in the HMM profile. The FP/FN trade-off plot further supports the selected threshold as expected. Overly stringent thresholds would exclude real Kunitz-domain proteins, likely including weaker or more divergent hits.

In the results it is also shown why the MCC was also more informative along with the accuracy rating. Because the negative dataset was much larger than the positive MCC provided a more balanced summary of performance by accounting for all four confusion matrix categories. Overall, the constructed profile HMM achieved reliable discrimination between Kunitz and non-Kunitz proteins.

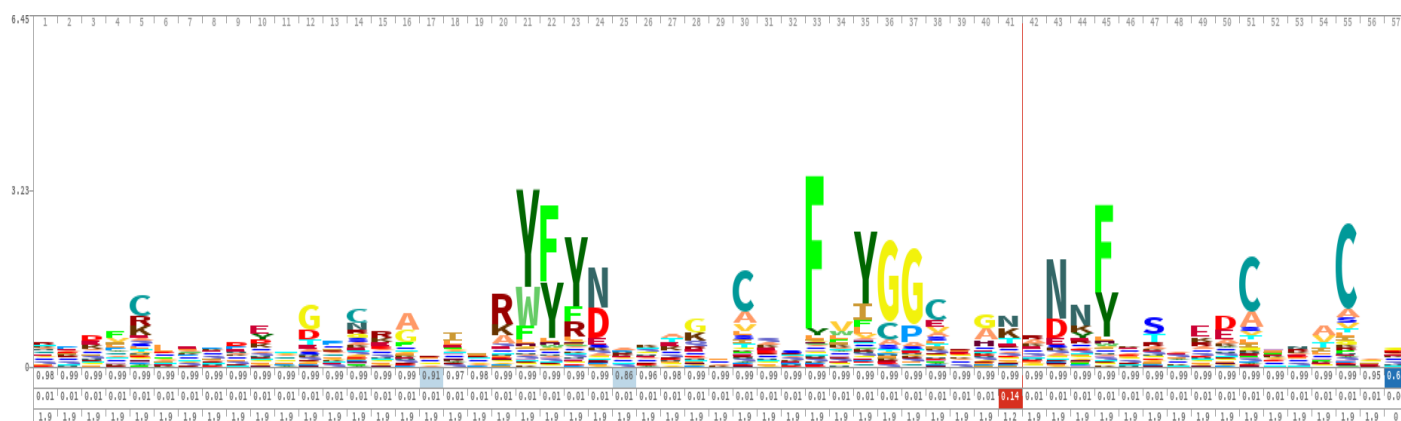


Figure 5. Skylign sequence logo of the final Kunitz profile HMM.

5. References

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6. Supplementary material

The final model, validation files, scripts, figures, and processed data are available in the accompanying GitHub repository, allowing the workflow to be reproduced and inspected.

<https://github.com/cloudy00000/kunitz-hmm-project>